Camp Fire Air Quality Analysis

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Project Overview

This project analyzes how the 2018 Camp Fire affected air quality in Northern California by comparing PM2.5 concentrations at EPA monitoring stations before and after the fire. The Camp Fire burned 153,336 acres in Butte County from November 8-25, 2018. The analysis uses a 100km buffer to identify affected monitors and includes a reusable ModelBuilder tool.

Key Findings:

- Average PM2.5 increase: **91.7 μg/m³ (+57%)**
- 10 monitors with complete data analyzed
- Impacts detected up to 100km from fire (Sacramento area)
- Regional air quality significantly degraded for weeks after fire

Research Question: *How did the Camp Fire affect PM2.5 concentrations at nearby monitoring stations?*

Data Sources

1. **Fire Perimeter**

Source: National Interagency Fire Center (NIFC)
Dataset: Historic Geomac Perimeters 2000-2018
URL: https://data-nifc.opendata.arcgis.com/
Details: Camp Fire, Nov 8-25 2018, 153,336 acres, Butte County CA

2. **Monitor Locations**

Source: U.S. EPA Air Quality System (AQS)
File: aqs_monitors.csv

URL: https://aqs.epa.gov/aqsweb/airdata/download_files.html
Filtered to: California (State Code 6), PM2.5 only (Parameter 88101)

3. **PM2.5 Measurements**

Source: U.S. EPA Air Quality System (AQS)
File: daily_88101_2018.csv (~2 million measurements nationwide)
URL: https://aqs.epa.gov/aqsweb/airdata/download_files.html
Filtered to: The period between October 9 - December 25, 2018 for California (State Code 6)

Data Processing:

- Created Monitor_ID field (format: 6-007-0008 = State-County-Site)
- Applied to both monitor locations and PM2.5 data for joining (Note: Somehow, the intermediate steps of joining monitor locations and PM2.5 data got deleted in Excel, and I couldn't recover them)
- Filtered to Before the fire (Oct 9-Nov 7) and After the fire (Nov 26-Dec 25) periods
- Excluded "During" fire period (Nov 8-25) due to extreme variability
- Final dataset: 10 monitors with complete before/after PM2.5data

Data Limitations:

- A limited number of selected monitors had complete data on PM2.5 readings

Methodology

a. The spatial analysis began by creating a 100km geodesic buffer around the Camp Fire perimeter using the Buffer tool. This buffer distance was chosen based on literature showing wildfire smoke can travel 50-200km depending on atmospheric conditions, allowing the analysis to capture both near-fire and regional downwind impacts.
b. Next, the Select Layer By Location tool identified all California PM2.5 monitors that intersected the 100km buffer zone, resulting in 27 selected monitors. These selected monitors were then exported to a permanent feature class using the Copy Features tool and saved as "Selected_Monitors." A Monitor_ID field was added to this output by combining State-County-Site codes (format: 6-007-0008) to enable joining with EPA measurement data.

a. *Data Processing (Excel)*

Monitor ID Creation: Combined State-County-Site codes into unique identifier:

Format: 6-007-0008 = California (6) - Butte County (007) - Site 8 (0008)

PM2.5 Data Filtering:

1. Filtered to California (State Code = 6)
2. Filtered to 27 selected Monitor_IDs
3. Filtered to Oct 9 - Dec 25, 2018 (77 days)
4. Result: 919 daily measurements

Time Period Definition:

- **Before Fire:** Oct 9 - Nov 7, 2018 (30 days baseline)
- **During Fire:** Nov 8-25, 2018 (excluded - extreme variability)
- **After Fire:** Nov 26 - Dec 25, 2018 (30 days post-fire)

Statistical Calculation:

- Created Excel Pivot Table (Monitor_ID × Period)
- Calculated averages for Before and After periods
- Derived: PM25_Change = After - Before
- Derived: Percent_Change = ((After - Before) / Before) × 100
- Final analysis: 10 monitors with complete data

C. Visualization (ArcGIS Pro)

Attribute Join: Joined monitor_statistics.csv to Selected_Monitors on Monitor_ID field. Exported to Monitors_Final.

Maps Created:

1. **Study Area Map:** Fire perimeter, 100km buffer, monitor locations
2. **Results Map:** Monitors symbolized by Percent_Change (graduated colors, 5 classes)

Chart: Bar chart with monitors showing impacts, side-by-side Before (blue) vs After (red) bars.

Results

Summary Statistics

- **Monitors analyzed:** 10 (with complete before/after data)
- **Average PM2.5 increase:** 91.7 μg/m³ (+57.5%)
- **Range of changes:** -61.6 to +245.9 μg/m³
- **Range of percent changes:** -15.2% to +145.4%

Key Findings

Magnitude of Impact: Average PM2.5 increase of 91.7 μg/m³ far exceeds EPA's 24-hour standard of 35 μg/m³. Nine of ten monitors showed increases, with largest impact being +246 μg/m³ (+98%). One monitor showed decrease (-15%), which could be due to upwind location.

Temporal Pattern: Elevated PM2.5 persisted for weeks after fire containment, indicating lingering smoke and atmospheric conditions.

Public Health Context: This magnitude of increase poses significant health risks for vulnerable populations (elderly, children, respiratory patients).

Tool Instructions

Fire Air Quality Analysis Tool

What It Does: Automates spatial selection of EPA monitors within specified distance of wildfire. This identifies which monitors to include in air quality impact analysis.

Prerequisites:

- Fire perimeter polygon (in your map)
- PM2.5 monitor locations (in your mapaa0

How to Run

Step 1: Open Tool Catalog -> Air_Quality_California.atbx-> Double-click "Fire Air Quality Analysis"

Step 2: Set Parameters

1. **Fire Perimeter:** Select fire polygon from dropdown
2. **PM2.5 Monitors:** Select monitor points from dropdown

3. **Buffer Distance:** Enter distance and unit (default: 100 km)
4. **Output Name:** Name for output feature class (e.g., Selected_Monitors)

After Running Tool

Tool performs spatial selection only. Next steps:

1. **Export Monitor IDs:** Open attribute table -> Copy Monitor_ID column -> Save as CSV
2. **Download PM2.5 Data:** Visit EPA site, download daily_88101_YYYY.zip
3. **Filter PM2.5 Data (Excel):** Create Monitor_ID, filter to your selected IDs and date range
4. **Calculate Statistics:** Define Before/After periods, create pivot table, calculate changes
5. **Visualize:** Join statistics to monitors in ArcGIS Pro, symbolize by change and create maps as necessary

<u>Post-Mortem</u>

What Went Well

a. **ModelBuilder Tool:** Tool works reliably. Creating parameters for buffer distance and output name adds real usability.
b. **Documentation Throughout:** Kept running notes as I worked. This saved hours when writing final documentation versus trying to remember everything at the end.

What Didn't Go as Planned

a. **Data Availability:** Only 10-12 of 27 selected monitors had complete PM2.5 data. Final analysis was based on fewer locations than expected.
b. **"During Fire" Period Decision:** Initially uncertain whether to include Nov 8-25 (active fire dates). Decided to exclude due to extreme variability. Those days aren't representative of steady-state conditions. "After" period shows lingering impacts, which is more policy-relevant.

Unexpected Discoveries

a. **Persistent Elevated PM2.5:** Even a month after fire containment, PM2.5 hadn't fully returned to baseline at some locations.

What I Would Do Differently

1. **Start with data availability check** - Download PM2.5 data first, filter to monitors with sufficient data, then do spatial selection
2. **Automate with Python** - Use Python to automate the data filtering and statistical calculations rather than manual Excel operations, making the workflow faster and repeatable for future fires

References

- U.S. EPA Air Quality System: https://aqs.epa.gov/
- NIFC Fire Perimeter Data: https://www.nifc.gov/
- EPA PM2.5 Standards: https://www.epa.gov/pm-pollution